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***Investigation and production of
eco-friendly materials by using
enhanced processing of husks
derived from Indian Arabica
and Robusta***

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ABSTRACT

The paper talks about the benefits of **byproducts of coffee production**, i.e. coffee grounds, and coffee husks as **raw material** for the manufacturing of **grocery bags and disposable cups**. The coffee grounds and coffee husks consist of a polysaccharide called **galactomannan** which can be obtained by two extraction methods, namely the water extraction method and an alkali extraction method. We draw attention to the **water extraction process** since it is **less complicated** and for its simplicity. Galactomannan must go through certain chemical processes for us to obtain **polylactic acid (PLA)**, which is a **biodegradable plastic** and hence can be used to make **eco-friendly products**. The paper further highlights a few case studies that show that sustainable products like this have had a major impact in other countries. Companies like **Huskee cups** and **Cuppa cookies** have implemented such products with coffee grounds, coffee husks, and edible cookies too, which is a sustainable alternative to traditional plastic products. Our goal is to present a **feasible and eco-friendly** solution to **reduce plastic** waste by **reusing coffee husks and grounds** into valuable materials for carrying groceries and as disposable cups.

Introduction

- Coffee husks and grounds
- Galactomannan
- Water Extraction Process
- Polylactic Acid
- Disposable Cups
- Carry Bags
- Sustainable



Problem statement

- Plastic Pollution Crisis
- Increase of Global Warming
- Methane Gas
- Coffee Byproducts



Literature Review

	Sl. no	Title	Author	Year of Publication	Review
	1	Sustainable Coffee Waste Management Through Circular Economy	Shatanik Chakraborty, Tushar Singh, Dhruv Agrawal, Aadil Ahmed Irshath and Anand Prem Rajan and Dr Anand Prem Rajan	2023	Water extraction method can be used to extract galactomannan from coffee grounds.
	2	TURNING TRASH INTO TREASURE: WASTE BANKS DRIVING CIRCULAR ECONOMY THROUGH COFFEE, PLASTIC, AND OIL WASTE MANAGEMENT	Gautama Sastra Waskita	2025	The byproducts of coffee end up in landfills where they decompose and release methane gas .
	3	The Innovative Swap System	Pontus Meier & Fanny Thulin	2019	The usage of PLA from coffee byproducts offers a solution to the global plastic pollution crisis
	4	EDIBLE COOKIE CUP: CUPPA COOKIE	Raja Nur Hanisah binti Raja Zainal Alam Shah	2021	Kaffeeform from Germany also uses a similar process using coffee grounds.

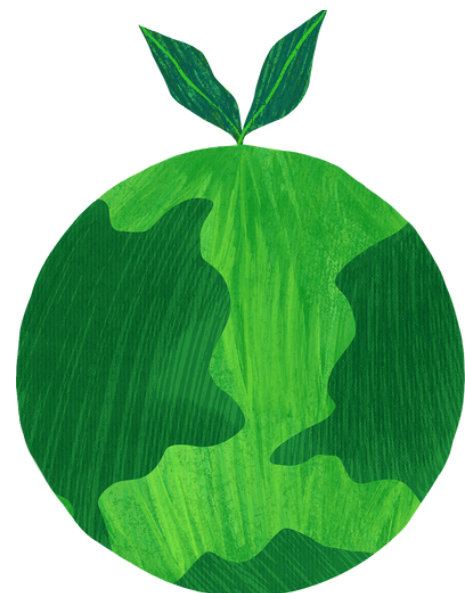
Literature Summary

This paper examines the use of coffee byproducts, like coffee grounds and husks, as raw materials for biodegradable products such as grocery bags and disposable cups. These byproducts contain galactomannan, which can be extracted using a simple water method and converted into polylactic acid (PLA), a sustainable alternative to plastic. Case studies of companies like Huskee cups and Cuppa cookies demonstrate the environmental benefits of this approach in reducing plastic waste.

S

STRENGTHS

- Sustainable
- Supporting Local Farmers
- Market Opportunities & Innovation



W

WEAKNESSES

- Technical Problems
- High Costs & Scalability
- Necessitating awareness



O

OPPORTUNITIES

- R&D Positions
- Waste Management
- Education & Advocacy



T

THREATS

- Technology Constraints
- Competition
- Regulatory Problems



Solution

- Repurposing of coffee grounds and husks
- Extraction of galactomannan
- Use of Polylactic Acid (PLA)
- Manufacturing Disposable cups and carry bags
- Sustainable
- Biodegradable



Reference

- [1] Waskita, Gautama Sastra. "TURNING TRASH INTO TREASURE: WASTE BANKS DRIVING CIRCULAR ECONOMY THROUGH COFFEE, PLASTIC, AND OIL WASTE MANAGEMENT." Vol. 3 No. 2 February 2025, page., 654-669
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- [3] Chakraborty, Shatanik, et al. "Sustainable Coffee Waste Management Through Circular Economy." (2023).
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[6] A Novel Approach to Utilize Used Disposable Paper Cups for the Development of Adsorbent and its Application for the Malachite Green and Rhodamine-B Des Removal from Aqueous solutions Kshipra Shukla, Alka Verma, Lata Verma, Shalu Rawat and Jiwan Singht† Department of Environmental Science, Babasaheb Bhimrao Ambedkar University, Lucknow-226025, India

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[11] <https://cupffee.me/en>



Thank You